TEXORA · LECTURE SHEET

Beat-up

How the reed, carried on the rocking sley, pushes the newly inserted pick home to the cloth fell: the crank-and-connecting-rod drive, sley eccentricity and its precious dwell at the back, beat-up force, and the cloth faults that follow when beat-up goes wrong.

TE-2103 · WEAVING TECHNOLOGY · TOPIC 8 · LEVEL 2 TERM 1

1. What beat-up is

Beat-up is the pushing of the newly inserted pick of weft to the **cloth fell** — the edge of the already-woven cloth, where the last beaten pick lies — by the **reed**. It is the third primary loom motion: shedding forms the opening, picking lays a loose pick across that opening at some distance from the fell, and beat-up closes the gap, packing the new pick against the previous picks so that the cloth gains its intended density, cover and firmness.

Without beat-up there would be no cloth, only a loose lattice. With it, each pick is driven into the V of the fell, where warp threads must bend around the new pick and the pick around them, locking the interlacing. The force rises steeply in the last few millimetres of travel — beat-up is a short, sharp effort at the front of the sley's swing, once per pick.

2. The sley (slay) and its parts

The **sley**, also spelt **slay**, is the rocking frame that carries the reed and the shuttle race and swings to and fro about a horizontal axis low in the loom. Its parts are:

- **Rocking shaft.** The horizontal shaft at the base of the loom about which the whole sley rocks. It is carried in bearings on the loom frame and is not itself driven round — it oscillates through a small angle as the connecting rods push and pull it.
- **Sley swords.** The two arms rising from the rocking shaft, one at each side, carrying race board and reed. They must be of equal length and square, or one reed end will beat harder and earlier than the other.
- **Race board / shuttle race.** The flat top of the sley on which the shuttle slides and the lower warp sheet rests. It must be smooth, level and exactly in line with the shuttle boxes.
- **Reed.** Fine parallel steel strips (dents) between two ribs, through which the warp ends pass. Its spacing sets warp density, its face guides the shuttle, and its edge delivers beat-up.

- **Reed cap (hand rail).** The heavy bar clamping the top of the reed. Its mass, far from the rocking axis, acts as a flywheel: its momentum carries the reed through the fell resistance and steadies the swing. Shuttle boxes sit at the race ends beneath it.

The sley is thus a timed, weighted pendulum: the race guides picking, the reed beats up, and the cap's mass supplies much of the beating energy — which is why changing cap weight or reed mass alters beat-up even when no setting has been touched.

3. The sley drive — crank and connecting rod

The sley is driven from the **crankshaft** — one revolution per pick. At each side a **crank** is joined by a **connecting rod** to a pin on the sley sword or rocking-shaft arm. As the crankshaft turns:

1. The crank pin travels in a circle at constant angular speed.
2. The connecting rod transmits it to the rocking shaft as push-and-pull.
3. The rocking shaft, swords, race and reed **rock to and fro** — forward for beat-up, back for shedding and picking.

If the connecting rod were infinitely long, the reed's motion would be simple harmonic. A real rod is short, and that shortness is deliberately used — the next section explains how.

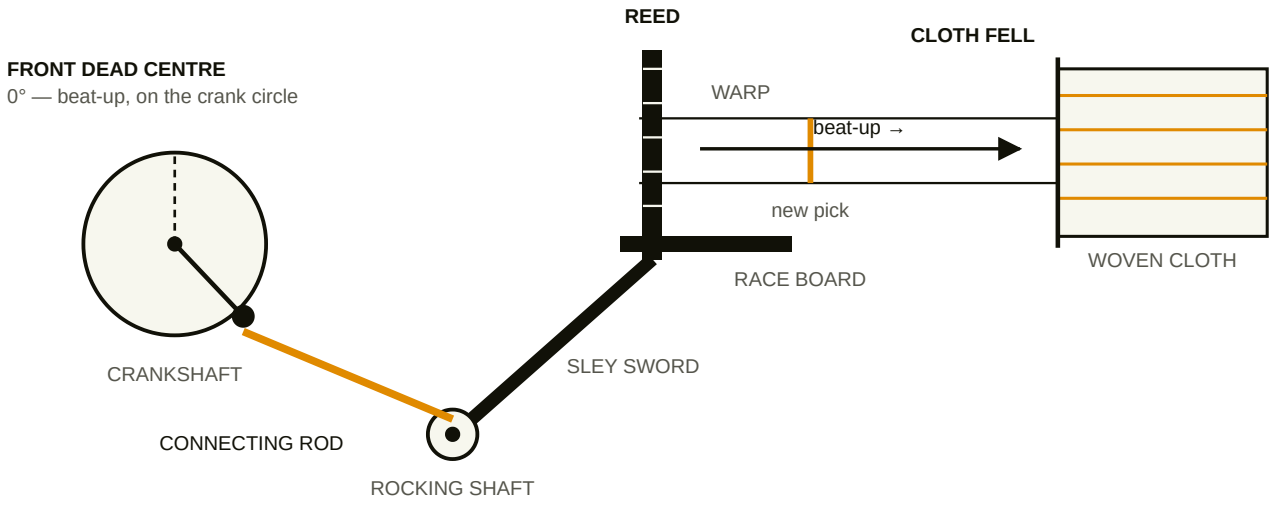


Figure 1 — Sley drive and beat-up. The crank on the crankshaft drives the rocking shaft through the connecting rod; the sley sword carries the reed forward so that at front dead centre (0°) the reed pushes the new pick to the cloth fell.

4. Sley eccentricity — the dwell that makes picking possible

4.1 Why the reed's motion is not simple harmonic

Because the connecting rod is short compared with the crank radius, the sley's angular motion is **not** symmetrical in time. At **front dead centre** crank and rod lie in line toward the sley pin (reed at the fell); at **back dead centre** they lie in line overlapped (reed furthest from the fell). The crank angles for the forward and return swings are *not* equal 180° shares: the sley spends **more crank angle — more time — in the back part of its swing and less in the front**. The reed **dwells near the back** and **moves quickly through beat-up**.

This asymmetry is called **sley eccentricity**, and it is exactly what weaving needs:

- **Dwell at the back** gives time — the scarcest quantity on a loom — for the shed to open fully and for the shuttle to fly across and be checked, with the race held steady beneath it (Topic 7: picking at 100–110°, boxed by about 220–240°).
- **Speed at the front** means the reed arrives at the fell with momentum and leaves again quickly, so the short, hard work of beat-up is done sharply and the reed does not linger pressing on the cloth and the open shed.

4.2 Eccentricity ratio

The degree of eccentricity is expressed by the ratio of crank radius to connecting-rod length:

$$e = \text{crank radius } (r) \div \text{connecting-rod length } (l)$$

A longer rod (small *e*) approaches simple harmonic motion, with little dwell. A shorter rod (large *e*) gives stronger back dwell and a sharper beat-up, but higher acceleration, vibration and drive strain. Typical looms use $e \approx 0.2–0.33$ (rod about three to five times the crank radius), a compromise between picking time and smoothness. Eccentricity is fixed by construction, but the weaver must know it: it sets the true picking window and the reed's velocity at the fell.

WORKED EXAMPLE 1 — ECCENTRICITY RATIO

A loom has a crank radius of 7.5 cm and a connecting rod 30 cm long. Find the eccentricity ratio and state its effect.

Step 1 — Formula: $e = r \div l$.

Step 2 — Substitute: $e = 7.5 \text{ cm} \div 30 \text{ cm} =$ **0.25**.

Step 3 — Interpretation: $e = 0.25$ lies in the usual range. The greater e is, the more the sley dwells near the back (more time for shedding and picking) and the sharper and quicker the movement through beat-up; the smaller e is, the closer the motion comes to simple harmonic, with less dwell and a gentler, more symmetrical swing.

WORKED EXAMPLE 2 — TIME FOR THE REED'S FINAL APPROACH

> A loom runs at 200 picks/min. The reed spends the last 60° of crank rotation before beat-up travelling to the fell. How long does that final approach take?
> Step 1 — Time for one crank revolution: 60 s ÷ 200 = **0.30 s per revolution**.
> Step 2 — The final approach is 60° of 360°, so time = (60 ÷ 360) × 0.30 s = (1/6) × 0.30 = **0.05 s**.
> Step 3 — Meaning: the whole resistance of the fell must be met and overcome in about one twentieth of a second. That is why beat-up depends on the momentum of the sley and reed cap, not on a steady push — there is no time for a steady push to act.

5. Beat-up timing

Beat-up occurs at **front dead centre, 0° (360°)** — crank, rod and sley pin in line, reed closest to the fell. Every other timing is referred to this instant: shedding changes while the reed is away from the fell, and picking (Topic 7) begins at about 100–110° after it.

In setting, the loom is turned by hand until the reed just touches the fell; the crank should then stand at front dead centre. Contact and dead centre are made to coincide by adjusting the drive connection, checking both sides: a square reed touches the fell along its whole width at the same instant. A reed touching at one side first is bent, wrongly capped, or carried by unequal swords — no timing adjustment will cure that.

The fell itself shifts pick by pick with take-up and let-off and can drift; since the reed's front position is fixed, a drifting fell changes the push distance and the force — several faults blamed on beat-up are really fell-position faults.

6. Beat-up force, picks per inch and warp tension

6.1 What the force must overcome

As the reed pushes the pick into the fell, three resistances grow together:

- **Warp tension.** To seat the pick, fell warp must deflect and stretch a little more; higher tension needs greater force and pushes the fell back more — the **fell retreat** that settles the pick at its true spacing as the reed withdraws.
- **Friction and bending.** Warp and weft must bend around each other into crimp form; fine, hairy or high-twist yarns resist more than smooth, soft yarns.
- **Proximity of previous picks.** Each pick enters a narrowing wedge; the higher the target **picks per inch (ppi)**, the steeper the force rise in the last fraction of travel.

6.2 Relation to picks per inch

The required force rises **sharply, not proportionally,** with picks per inch. At low ppi the pick slides home easily. As ppi approaches the maximum the yarn diameters allow — the *jammed* condition, warp and weft already in full contact — the force for each further small increase grows very large, the sley deflects, warp breaks rise and the cloth turns boardy. In practice take-up is set for a ppi comfortably below the jammed limit. Division of labour: **take-up decides the nominal ppi; beat-up must supply the force to make it real.** If force is insufficient, spacing opens out, the fell retreats, and the cloth weaves lighter than the setting promises.

6.3 Relation to warp tension

Raising warp tension gives a firmer fell and crisper beat-up, but raises the force needed and warp breakage, and reduces weft crimp so cloth may look thin despite high ppi. Lowering tension eases beat-up and improves cover, but a slack warp gives a wandering fell and uneven spacing. Beat-up force, warp tension and take-up form a triangle: change one and re-check the other two. Weavers judge the balance by the sound of beat-up — a clean, firm note, equal at both sides — and by the woven ppi measured off the loom.

7. Faults of beat-up and their interplay with take-up

FAULT	APPEARANCE IN CLOTH	BEAT-UP / TAKE-UP CAUSES
Reed marks (reedy cloth)	Faint lengthwise lines or stripes in the cloth, following the reed dent spacing; warp ends grouped visibly dent by dent instead of lying evenly.	Damaged, bent or dirty dents; wrong reed count; uneven warp per dent. Strictly a reed fault, printed by beat-up force.
Thick places / bars across the cloth	A band where picks are packed too closely for a few picks — a dark, dense line across the width, often at a loom restart.	Fell moved toward the reed while the loom stood (warp relaxing / cloth creeping), so the first beats after restarting pack picks tightly; take-up not moving in step with the first picks.
Thin places / cracks	A band where picks are too far apart — a light, open line across the width.	Fell fallen away from the reed (take-up pulling cloth on while the loom stood, or warp slipping), weak beat-up after restart, or take-up running fast relative to picking.
Uneven ppi across the piece	Alternating light and heavy sections along the length, not single bars.	Irregular beat-up force (worn crank pins, loose rods, unequal swords) combined with irregular take-up (worn ratchet, slipping wheel).

Starting marks (general)	Any thick or thin bar at the exact place the loom was stopped and restarted.	The classic beat-up / take-up interplay fault: during the stop, fell position changes; on restart, beat-up force and take-up do not resume in their running relationship for the first few picks.

Beat-up does not act alone. The reed can only push a pick to wherever the fell is; take-up and let-off decide where the fell is. Most cross-bars are diagnosed by asking together: where was the fell, how hard did the reed beat, and how much cloth did take-up draw away on those picks?

Glossary

Beat-up — pushing the new pick to the fell with the reed. **Cloth fell** — edge of woven cloth where the last beaten pick lies. **Sley (slay)** — rocking frame carrying reed and race. **Sley sword** — side arm of the sley. **Rocking shaft** — shaft about which the sley oscillates. **Race board** — flat sley top on which the shuttle slides. **Reed cap** — heavy bar clamping the reed; its mass aids beat-up. **Crankshaft** — main shaft, one revolution per pick. **Connecting rod** — rod joining crank to sley. **Front dead centre** — crank position (0°) of beat-up. **Sley eccentricity** — time-asymmetry of the swing from a short rod: dwell at back, speed at front. **Eccentricity ratio** — e = crank radius ÷ rod length. **Picks per inch (ppi)** — weft picks in one inch of cloth. **Reed mark** — lengthwise line from a damaged reed. **Starting mark** — thick or thin bar at a loom restart.

Key points

• Beat-up pushes each new pick to the cloth fell with the reed, once per crank revolution, at front dead centre (0°).
• The sley — rocking shaft, swords, race board, reed and reed cap — is driven from the crankshaft by a crank and connecting rod at each side.
• A short connecting rod makes the motion eccentric: the reed dwells near the back (time for shedding and picking) and moves quickly through beat-up.
• Eccentricity ratio e = r ÷ l; typical values are about 0.2–0.33. Greater e gives more dwell and a sharper beat, at the cost of vibration and strain.
• Beat-up force rises sharply with picks per inch and with warp tension; take-up sets the nominal ppi, beat-up must supply the force to achieve it.
• Reed marks come from the reed itself; thick and thin places and starting marks come from the interplay of beat-up force, fell position and take-up.

Practice

1. A loom has a crank radius of 6 cm and a connecting rod of 24 cm. Calculate the eccentricity ratio and state whether its dwell at the back is greater or less than that of a loom with e = 0.20.
2. A loom runs at 240 picks/min. Calculate the time taken for the last 60° of crank rotation before beat-up, showing every step.
3. Name the five parts of the sley listed in this sheet and state the function of each in one line.
4. Explain why the reed does not move with simple harmonic motion, and give the two practical benefits of the motion it actually has.
5. A cloth shows a thin bar across its width every time the loom is restarted. Explain the cause in terms of fell position, beat-up and take-up, and state two remedies.
6. Why does the beat-up force rise sharply — not proportionally — as picks per inch approach the jammed condition? Answer in terms of what the pick must do at the fell.

Answer key

1. e = r ÷ l = 6 ÷ 24 = **0.25**. Since 0.25 > 0.20, this loom has the **greater dwell** at the back and the sharper beat-up.
2. One revolution: 60 ÷ 240 = 0.25 s. Time for 60° = (60 ÷ 360) × 0.25 = (1/6) × 0.25 ≈ **0.042 s** (0.0417 s).
3. Rocking shaft — the rocking axis. Swords — arms carrying race and reed. Race board — shuttle guide surface. Reed — sets warp spacing and beats up. Reed cap — clamps the reed; its mass supplies momentum.
4. The connecting rod is short relative to the crank radius, so the crank angles for the forward and return halves of the swing are unequal; the motion is eccentric, not simple harmonic (which would need an infinitely long rod). Benefits: (1) dwell near the back gives time for shedding, picking and checking; (2) rapid movement at the front gives a sharp beat-up with momentum and clears the reed quickly from the fell.
5. At a stop, warp and cloth relax and the fell creeps from its running position. On restart the first picks are beaten against a displaced fell, with beat-up and take-up out of step, so they space too widely — a thin bar. Remedies (any two): restore the fell to its running position before restarting; beat the first picks with take-up eased or held; check tension and fell position at every stop.
6. The pick must force into a narrowing wedge, bending warp around itself against friction and tension. Near the jammed spacing threads are already in full contact, so each further reduction needs a large extra deflection of every thread — resistance climbs steeply, not proportionally.
